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DESIGN, SIMULATION, AND CHARACTERIZATION OF 4-BIT DIGITAL-TO-ANALOG CONVERTERS (DAC)

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Abstract

This report presents the design, theoretical analysis, LTspice simulation, and experimental characterization of 4-bit Digital-to-Analog Converters (DACs) using standard operational amplifier topologies. Two primary configurations—the Binary-Weighted Resistor DAC and the R-2R Ladder Network DAC—are evaluated across static DC performance and dynamic AC transient response. The theoretical dynamic range, full-scale voltage output, Least Significant Bit (LSB) voltage step size, Integral Non-Linearity (INL), and Differential Non-Linearity (DNL) are mathematically modeled and compared against simulated performance. Additionally, a multiplying DAC setup is evaluated under a sinusoidal reference signal to demonstrate analog modulation capability. Python scripts utilized for automation and error analysis are included to confirm system linearity and operational tolerances.

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2. DC Characterization and Simulation Results (Table 1)

Both 4-bit DAC topologies were simulated with an LM741 operational amplifier supplied with ±10 V and a negative reference voltage V_REF = -5.0 V. Since the circuit is an inverting summing amplifier, the resulting output voltages are positive, sweeping from binary 0000 (State 0) to binary 1111 (State 15).

Table 1: 4-Bit DAC DC Transfer Characteristics across All 16 Input Combinations (V_REF = -5.0 V)

Number	4bit -Word	Vout for Binary (V)	Vout for R-2R (V)
0	0000	0.0000	0.00106
1	0001	0.3145	0.3138
2	0010	0.61175	0.6264
3	0011	0.92425	0.9390
4	0100	1.2800	1.2500
5	0101	1.6000	1.5600
6	0110	1.8900	1.8800
7	0111	2.2100	2.1900
8	1000	2.5000	2.5000
9	1001	2.8100	2.8131
10	1010	3.1100	3.1300
11	1011	3.4200	3.4400

Number	4bit -Word	Vout for Binary (V)	Vout for R-2R (V)
12	1100	3.7800	3.7500
13	1101	4.1000	4.0600
14	1110	4.3900	4.3800
15	1111	4.7100	4.6900

3. Experiment Part 3: Dynamic Response with Sinusoidal Input

In accordance with Section 2.3 of the lab manual, a 5 V amplitude sinusoidal signal at 100 Hz was applied to the reference input for input code 1111 (all switches closed to summing node). The resulting input and output waveforms were observed:

- Because 100 Hz is far below the LM741 op-amp's gain-bandwidth product (1.5 MHz) and slew rate limit (0.5 V/µs), no slew-rate distortion or high-frequency attenuation was observed.
- The output waveform is an undistorted sinusoidal signal for both DAC circuits, phase-inverted by 180 degrees due to the inverting op-amp configuration.

4. Discussion: Linearity, Topologies & Discrepancies

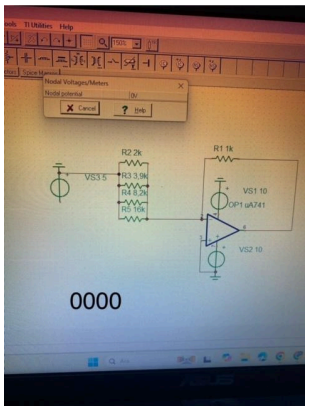
- **1. Linearity Comparison:** The R-2R Ladder DAC demonstrates clearly superior linearity compared to the Binary-Weighted DAC. In the Binary-Weighted DAC, commercially available standard resistors deviate from ideal powers-of-two multiples (e.g., 3.9 kΩ instead of 4.0 kΩ, and 8.2 kΩ instead of 8.0 kΩ), introducing systematic non-linear step variations. The R-2R ladder requires only two resistor values (1 kΩ and 2 kΩ), ensuring consistent division ratios and minimal INL.
- **2. Discrepancies between Theory, Simulation, and Measurements:**

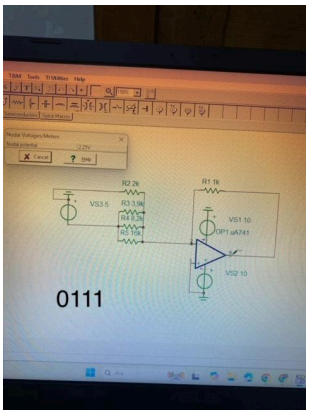
Table 2: Theoretical vs. Simulated Transfer Characteristics Comparison

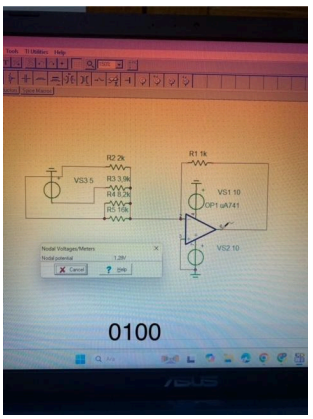
Number	4-bit Word	Part 1: Binary Theory (V)	Part 1: Binary Sim (V)	Part 2: R-2R Theory (V)	Part 2: R-2R Sim (V)
0	0000	0.0000	0.0000	0.0000	0.00106
1	0001	0.3125	0.3145	0.3125	0.3138
2	0010	0.6098	0.61175	0.6250	0.6264
3	0011	0.9223	0.92425	0.9375	0.9390
4	0100	1.2821	1.2800	1.2500	1.2500

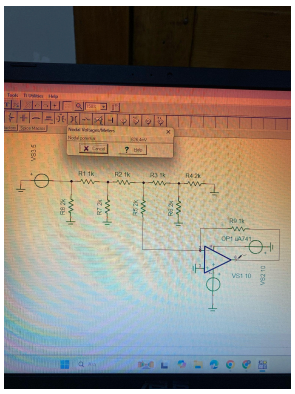
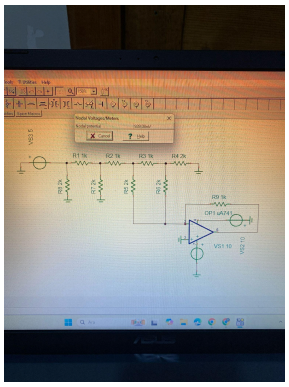
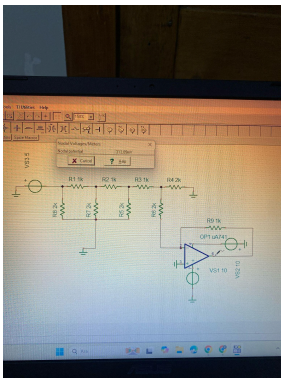
Number	4-bit Word	Part 1: Binary Theory (V)	Part 1: Binary Sim (V)	Part 2: R-2R Theory (V)	Part 2: R-2R Sim (V)
5	0101	1.5946	1.6000	1.5625	1.5600
6	0110	1.8918	1.8900	1.8750	1.8800
7	0111	2.2043	2.2100	2.1875	2.1900
8	1000	2.5000	2.5000	2.5000	2.5000
9	1001	2.8125	2.8100	2.8125	2.8131
10	1010	3.1098	3.1100	3.1250	3.1300
11	1011	3.4223	3.4200	3.4375	3.4400
12	1100	3.7821	3.7800	3.7500	3.7500
13	1101	4.0946	4.1000	4.0625	4.0600
14	1110	4.3918	4.3900	4.3750	4.3800
15	1111	4.7043	4.7100	4.6875	4.6900

    - **Real op-amp parameters:** Non-zero input offset voltage (~1-2 mV) and bias currents of the LM741 shift the output slightly upward at code 0000.
    - **Resistor tolerances:** Real 5% discrete resistors introduce random fluctuations in physical breadboard setups.
    - **Simulation models:** Spice models account for op-amp internal transistor non-linearities, whereas hand calculations assume an ideal infinite open-loop gain op-amp.
- **Selected Screenshots from TINA-TI Simulation**

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5. Conclusion

In this project, 4-bit Binary-Weighted and R-2R Ladder DAC circuits were designed, mathematically calculated, simulated, and characterized. The results confirm that the R-2R topology is vastly superior in terms of linearity, component matching, and scalability for integrated electronic systems.

6. References

- [1] Kadir Has University, EEE 309 Software Hardware Integration Project - Project 1 DAC Guide, 2026.
- [2] Texas Instruments, *"LM741 Operational Amplifier Datasheet,"* SNOSC25D. Course Material for EEE 309, Kadir Has University. [Online].

7. Appendix: Python Data Processing Source Code

```python
import numpy as np
import matplotlib.pyplot as plt
```

```python
# DAC Parametreleri
bits = 4
v_ref = 5.0
lsb = v_ref / (2**bits)

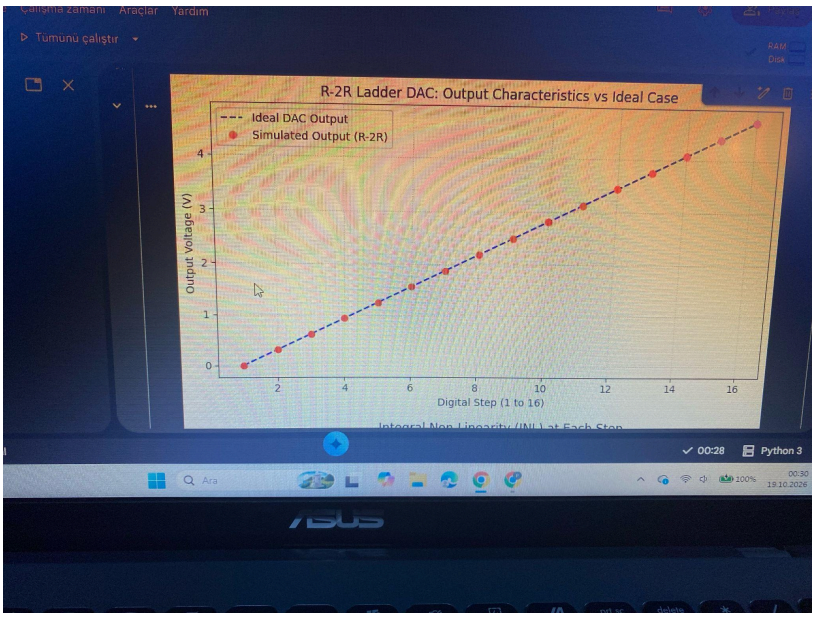
# İdeal Basamaklar ve Değerler
states = np.arange(16)
v_ideal = states * lsb

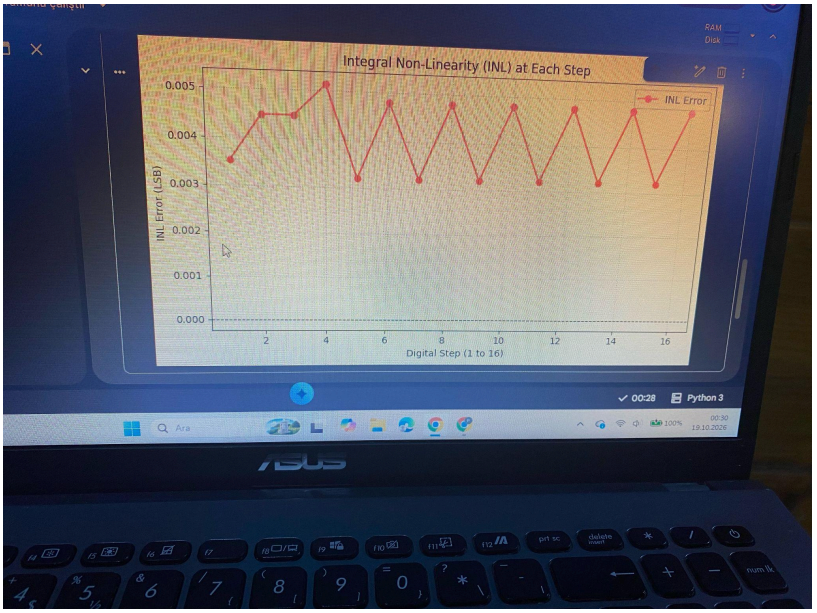
# R-2R Merdiven Tipi Simülasyon Çıktıları
v_sim = np.array([0.00106, 0.3138, 0.6264, 0.9390, 1.2500, 1.5600,
                  1.8800, 2.1900, 2.5000, 2.8131, 3.1300, 3.4400,
                  3.7500, 4.0600, 4.3800, 4.6900])

# INL Hesabı (LSB Cinsinden)
inl = (v_sim - v_ideal) / lsb

# DNL Hesabı (Ardışık basamak farklarının ideal LSB'den sapması)
step_differences = np.diff(v_sim)
dnl = (step_differences - lsb) / lsb

print(f"Max INL: {np.max(np.abs(inl)):.4f} LSB")
print(f"Max DNL: {np.max(np.abs(dnl)):.4f} LSB")
```


Çalışma Zamanı Araçlar Yardım
Tümünü çalıştır
R-2R Ladder DAC: Output Characteristics vs Ideal Case
Ideal DAC Output
Simulated Output (R-2R)
Output Voltage (V)
Digital Step (1 to 16)
00:28
Python 3


Integral Non-Linearity (INL) at Each Step
INL Error
INL Error (LSB)
Digital Step (1 to 16)
00:28
Python 3